

DoWaun

Passive Air Curtain: HGV Trailer Skirt Integration

CFD-Led Aerodynamic Study

EG6166: Mental Wealth Professional Life

Technical Report

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Abstract

Heavy goods vehicle (HGV) trailers are a significant source of aerodynamic drag on public roads, with the exposed bogie assembly beneath the rear overhang being a particularly turbulent and poorly managed flow region. This report presents the design and computational investigation of a passive air curtain device integrated within the trailer side skirt, intended to deliver a controlled sheet of air at the bogie and reduce pressure drag beyond what a conventional skirt achieves. The device draws on the aerodynamic principles of automotive wheel air curtains, channelling dynamic pressure from a side-facing intake (850 mm × 850 mm at 18.5° to the flow direction) through a 5 m duct and discharging it through a 50 mm × 400 mm slot. Four configurations are evaluated using ANSYS Fluent with a steady-state RANS k- ω SST approach: a baseline (no skirt), a skirt-only case, and two air curtain variants comprising a wide curved outlet and a thin narrow slot. Simulations are conducted at yaw angles of 0°, 7.5°, and 15° in a domain with a blockage ratio of 1.25%, well within accepted limits. Preliminary analysis indicates that, at zero and moderate yaw, the leading skirt intake region receives limited positive pressure; however, at 15° yaw a pressure differential of approximately 1,200 Pa develops, providing a strong passive driving force. Initial results from the narrow slot design show backflow within the duct under certain conditions, highlighting the sensitivity of the pressure budget to geometry and yaw angle.

Table of Contents

Abstract	2
Table of Contents	3
1. Introduction	4
1.1 Aim and Objectives.....	4
2. Literature Review	5
2.1 Existing Aerodynamic Devices	5
2.2 Yaw Effects on HGV Aerodynamics	5
2.3 Passive Flow Control and Pressure Budget.....	6
3. Methodology	7
3.1 Concept Designs	7
3.2 Design Analysis	12
3.3. CFD Setup	14
3.3.1 Mesh and Boundary Conditions.....	16
3.3.2 Solver Settings	Error! Bookmark not defined.
3.4 CFD Test Configurations	Error! Bookmark not defined.
3.5 System Integration Approach	16
4. Results and Discussion	17
4.1 Intake Pressure Distribution.....	Error! Bookmark not defined.
4.2 Surface Pressure Distribution	Error! Bookmark not defined.
4.3 Design Comparison	17
4.4 Limitations	22
5. Conclusion	23
4. Methodology	Error! Bookmark not defined.
6. References	29
7. Appendix.....	30

1. Introduction

Heavy goods vehicles (HGVs) are a major source of global CO₂ emissions. In the United Kingdom, HGVs account for approximately 20% of all road transport carbon dioxide output despite being a small fraction of total vehicle numbers. At typical motorway operating speeds of 56 MPH, aerodynamic drag is the dominant resistive force on a HGV, accounting for approximately 25–30% of total fuel consumption (Cooper, 2003). NACFE (2016) reports that aerodynamic technologies deliver measurable fuel savings, and these benefits scale significantly when applied across large national or international fleets.

The HGV trailer presents a particularly challenging aerodynamic problem. Unlike passenger cars, which have been extensively optimised using wind tunnel testing and computational fluid dynamics (CFD), most trailers have box-shaped structures with sharp rear edges and an exposed bogie assembly beneath the rear overhang. The rear wheel assembly is a primary source of underbody drag. As airflow passes beneath the trailer, the rotating wheels and exposed suspension components cause early flow separation, producing a large, turbulent wake. This unsteady low-pressure region limits pressure recovery at the rear of the vehicle and increases the overall aerodynamic resistance.

Trailer side skirts are already a widely deployed countermeasure. By extending the trailer sidewall toward the ground, the skirt prevents external flow from entering the underbody region and interacting with the wheelset. NACFE (2024) documents typical fuel savings of 1–5% for skirted trailers relative to unskirted equivalents. However, conventional skirts act by blocking flow rather than by actively shaping or managing the residual flow that reaches the axle area.

This report outlines the theoretical basis, design geometry, computational methodology, and preliminary findings of the study. Final quantitative results will be incorporated upon completion of all CFD runs. Four configurations; baseline with no skirt, baseline with skirt, and two modified air curtains, are evaluated using ANSYS Steady State RANS CFD at yaw angles of 0°, 7.5°, and 15°, the latter two reflecting measured UK on road conditions (Gerova & Barden, 2016). The study aims to quantify any drag reduction beyond the skirt alone and to identify the geometric and yaw dependent factors that govern performance.

1.1 Aim and Objectives

The aim of this project is to design, simulate, and critically evaluate passive air curtains designs integrated within an HGV trailer side skirt that reduces aerodynamic drag at the bogie, using CFD under representative on road yaw conditions.

Objectives:

1. Develop a conceptual design for a skirt mounted air curtain, specifying intake geometry, internal duct routing, and alternative outlet configurations.
2. Construct CAD models and perform steady state RANS CFD simulations for four configurations at 0°, 7.5°, and 15° yaw.
3. Quantify drag coefficient changes and compare the aerodynamic performance of the wide curved outlet and thin narrow outlet designs.

4. Critically evaluate the results in the context of pressure budget theory, identify practical design trade-offs, and consider the professional implications including safety, legal compliance and sustainability.

2. Literature Review

2.1 Existing Aerodynamic Devices

Wheel air curtains are now common in modern passenger vehicle aerodynamics. BMW describes its air-curtain system as front bumper channels that guide oncoming air along and around the wheels, reducing the turbulence created at the tyre contact patch and inside the wheel arch (BMW, n.d.). This supports the present project by providing industry validated evidence. Interventions that control airflow around the wheel area such as skirts, ducts and guiding surfaces, have a measurable impact on reducing separation and improving aerodynamic efficiency. The BMW example therefore strengthens the theoretical basis for investigating wheel drag reduction concepts in this study.

Hyundai and Kia's Active Air Skirt on the Genesis GV60 routes high-pressure air ahead of the front wheels and reduces the drag coefficient by 0.008 (approximately 2.8 %) (Hyundai Motor Company, 2024). Both systems rely on the vehicle's own dynamic pressure, confirming that passive air curtains are industrially viable. Furthermore, it shows that wheel region airflow can be controlled using only the vehicles natural dynamic pressure, demonstrating passive air curtain concepts are already proven in industry practice.

NACFE identifies trailer side skirts as the most effective commercial underbody aerodynamics drag reduction technology. Their research notes that the physical mechanism works by blocking the lateral gap between the trailer floor and the road surface, to prevent high-momentum external flow from entering the underbody and striking the wheelset directly (NACFE, 2015). This demonstrates that controlling the airflow around the wheel region is a proven method for reducing turbulence and drag, supporting the relevance of investigating similar flow management concepts in this study.

Wind-tunnel studies on the DrivAer model further support the potential for wheel region flow control. (Nabutola & Boetcher, 2021) showed that both conventional and air-jet wheel deflectors can alter the wheel-housing boundary-layer interaction, producing measurable aerodynamic drag reductions. These studies, however, are limited to light-duty vehicles, and no published work has examined whether a passive air curtain concept could be applied to the exposed bogie region of a heavy goods vehicle. This gap highlights the need for the present study.

2.2 Yaw Effects on HGV Aerodynamics

Selecting representative yaw angles for aerodynamic evaluation requires evidence of actual on-road conditions. Barden and Gerova (2016) instrumented a UK tractor trailer over two months of representative delivery routes and reported an average absolute yaw angle of 7.9° , giving an effective yaw of approximately 7.1° . Strong crosswinds occasionally produced angles exceeding 15° . The present study therefore evaluates the air curtain at 7.5° (representative average) and 15° (extreme crosswind) to capture both typical and demanding operating conditions. The distribution of on-road yaw angles is shown in Figure 2.2.1.

J. Barden, K. Gerova / Transportation Research Part D 48 (2016) 284–297

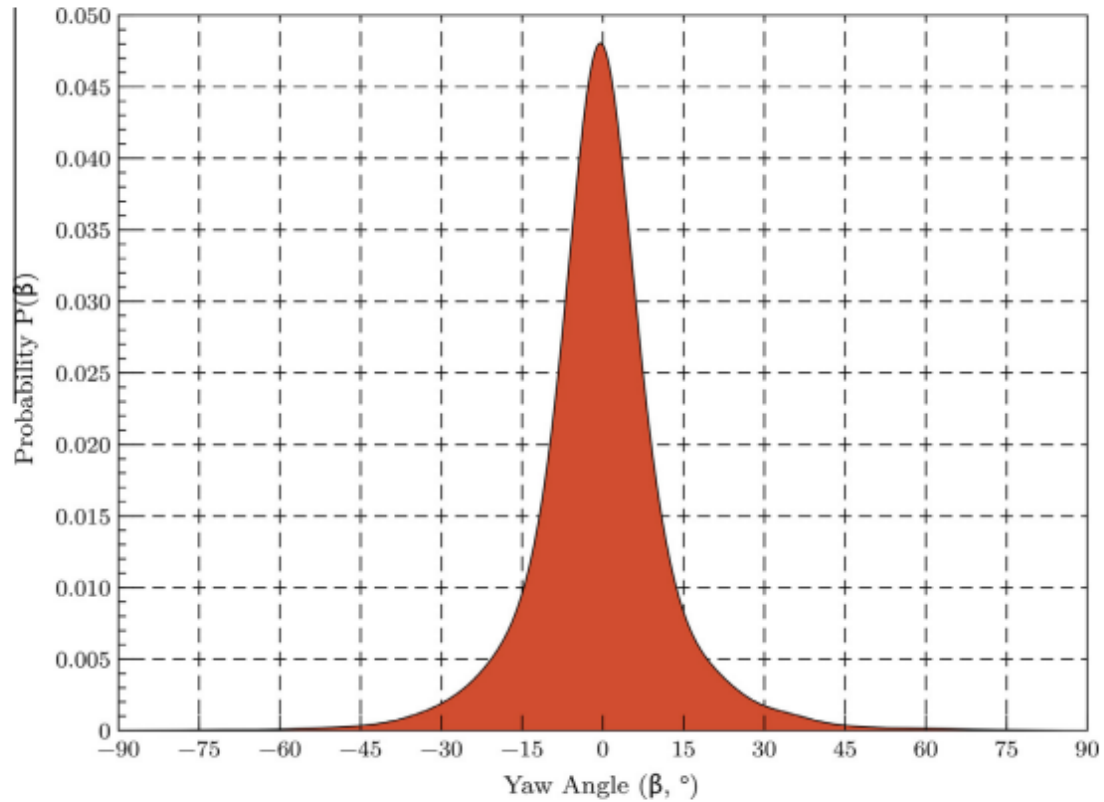


Fig. 7. Wind yaw angle probability ($P(\beta)$) vs. yaw angle.

Figure 2.2.1: Wind yaw angle probability distribution (Barden & Gerova, 2016).

Based on these findings, the present study evaluates the air curtain concept at 7.5° to represent typical operating conditions and at 15° to capture more severe but plausible crosswind scenarios. This approach ensures that both typical and extreme aerodynamic environments are reflected in the analysis.

2.3 Passive Flow Control and Pressure Budget

Passive flow control devices rely entirely on the vehicle's own motion for their driving force, drawing energy from the available dynamic pressure rather than requiring external power input. Their effectiveness is therefore determined by the available dynamic pressure at the intake and the internal losses through the delivery system; a pressure budget approach. Their performance ceiling is set by the available dynamic pressure in the freestream:

$$q = 0.5\rho U^2 \quad (1)$$

Where:

- q is the dynamic pressure (typically measured in Pa or N/m^2),
- ρ is the Fluid Density ($\approx 1.225 \text{ kg/m}^3$), $\approx$ sea-level air density),

- U is vehicle speed.

While the concept of pressure budgeting is a straightforward application of fluid mechanics principles its systematic application to a skirt integrated air curtain has not been reported. (White, 1999).

Existing passive HGV devices, such as skirts and base bleed systems, capture high pressure air but do not combine a side skirt intake with a ducted air curtain outlet, nor do they explicitly optimise the system as a pressure budgeted design across a yaw sweep (Transport Canada, 2012). This study addresses that gap by integrating an angled intake, a 5 m internal duct, and two outlet geometries into a trailer skirt, and evaluating performance using a pressure budget framework interpreted through steady state RANS CFD. The approach provides both a practical device concept and a systematic method for analysing the trade-offs between intake location, duct geometry, and outlet design.

3. Methodology

3.1 Concept Designs

For this study, the methodology includes two baseline design studies. Two parametrically modified concept designs are chosen for aerodynamic analysis. The baseline design models represent a standard HGV truck configuration with and without trailer skirt installed. The purpose of the baseline models is to serve as control cases for assessing the aerodynamic performance before modification is introduced. The results from these cases are then compared with the modified concept designs with improved trailer skirts to evaluate the effects of the skirt on drag reduction and airflow behaviour. The baseline models will identify, the drag coefficient, pressure distribution and wake formation. These results are used to evaluate the effectiveness of a trailer skirt. **Figure 3.1.1** and **Figure 3.1.2** presents the baseline HGV models used as control cases.

Table 3.1.1: Trailer Aerodynamic Test Configurations

1	Baseline 1	Standard defeatured tractor-trailer with no aerodynamic add-ons. Reference drag-coefficient for all comparisons.
2	Baseline 2	Conventional trailer side skirt fitted, no air curtain. Quantifies drag reduction attributable to the skirt alone.
3	Concept 1 (Thin Narrow Slot)	Full air curtain system with thin, narrow outlet slot. Tests whether a high-momentum concentrated jet is more effective.
4	Concept 2 (Wide Curved Slot)	Full air curtain system with wide, curved outlet slot. Tests whether a broad, guided discharge reduces bogie drag.

Baseline Design 1

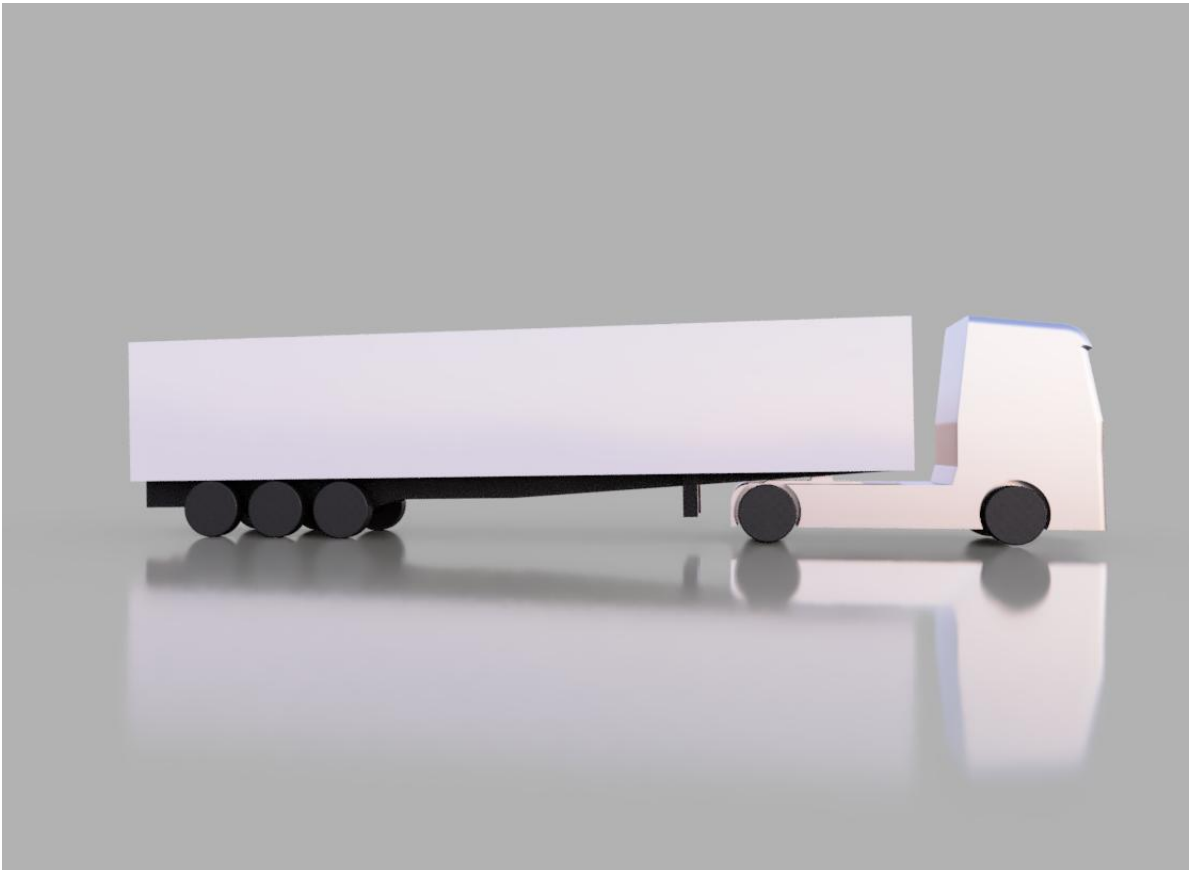


Figure 3.1.1: Side View of Baseline 1 HGV Truck Without Trailer Skirts

Baseline design 1 consists of a standard HGV tractor unit and trailer without skirts. In this design, the underside, wheels and axles are fully exposed, allowing air to freely pass beneath the trailer. This is the most common trailer configuration. The units together follow typical bluff body flow behaviour. This is used as an absolute baseline.

Baseline Design 2

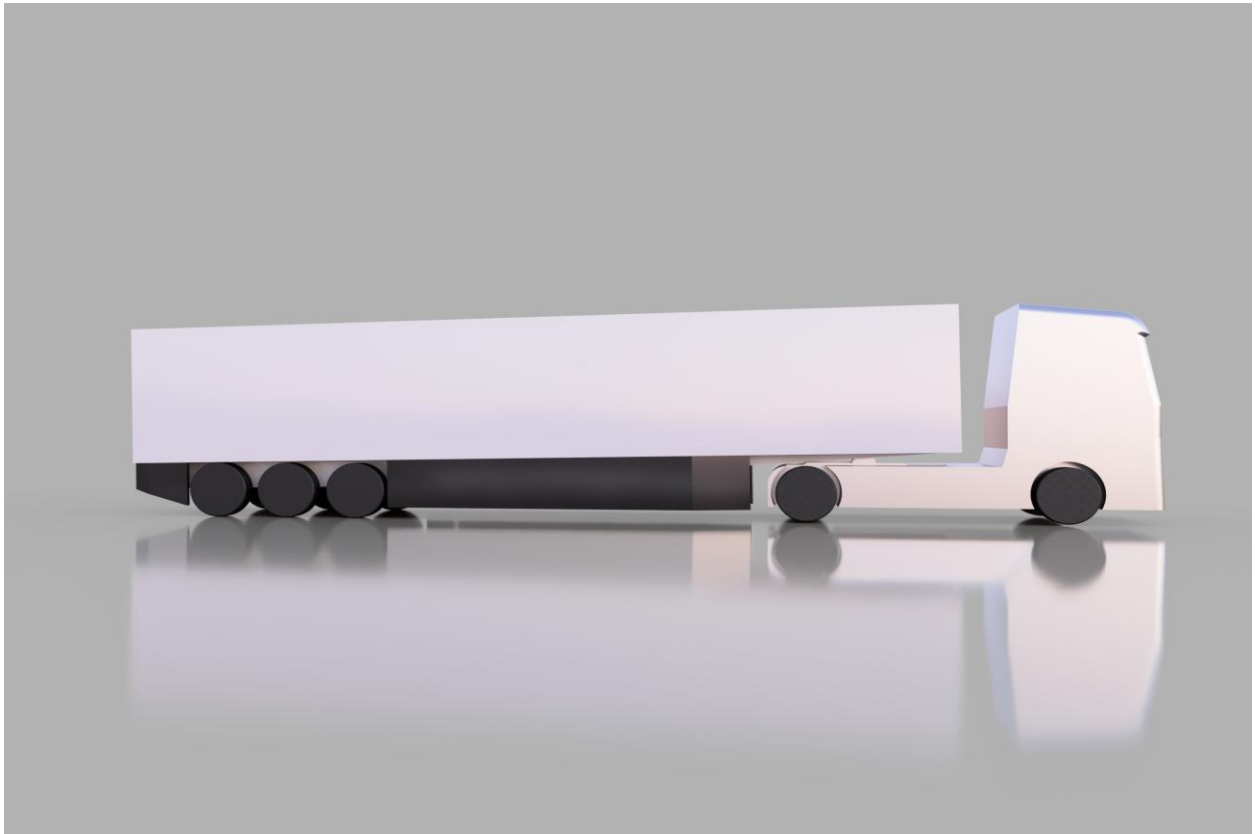


Figure 3.1.2: Side View of Baseline 2 HGV Truck with Unmodified Trailer Skirts

Baseline design 2 represents a standard HGV tractor with trailer that has a typical skirt and ducktail. This design is used as a reference case to compare the aerodynamic performance of a conventional trailer skirt before introducing further design changes, such as holes or modified geometry.

Concept Design 1

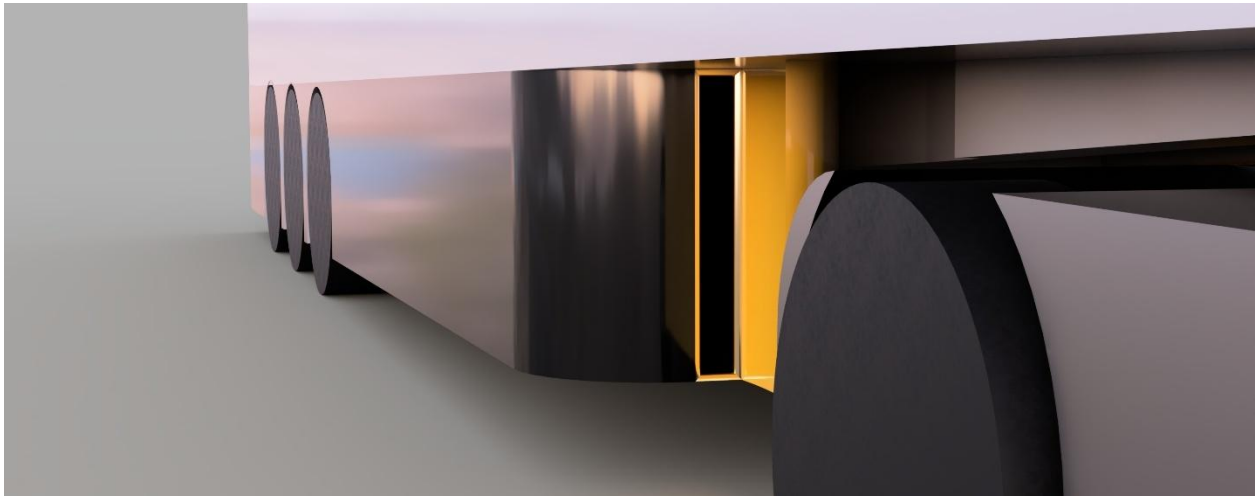


Figure 3.1.3: *Detailed View of Trailer Skirt Inlet Geometry Design 1*

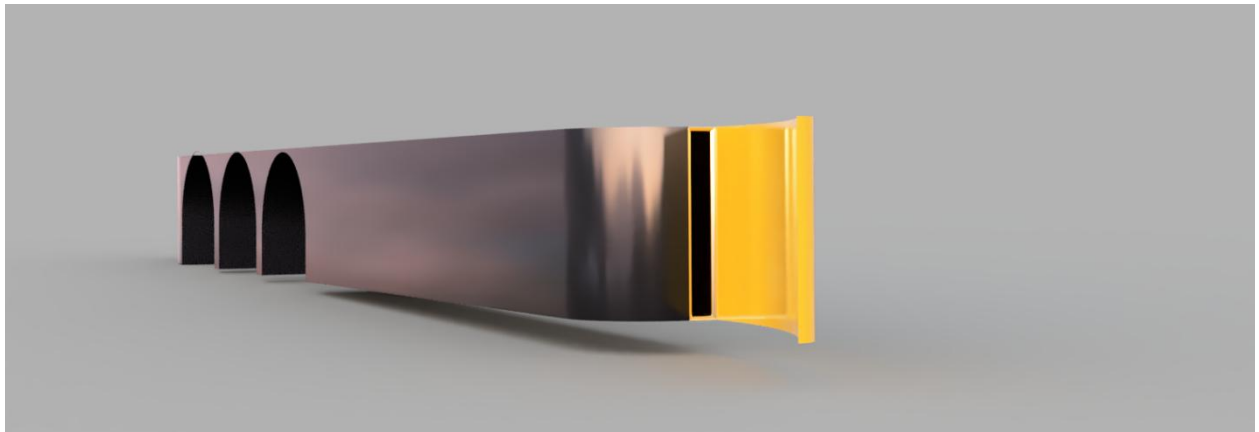


Figure 3.1.4: *Isolated View of Trailer Skirt Inlet Geometry Design 1*



Figure 3.1.5: *Isolated View of Trailer Skirt Outlet Geometry Design 1*

Concept design 1 consists of a trailer skirt with scoop capture device leading to a thin slot. This modified skirt is fitted along the lower side of the trailer, positioned between the trailer wheels and underside of the trailer body. The purpose of this design is to capture the high-pressure air at the skirt face then guide it, with the scoop, into the slot. This narrows, allowing a jet to form across the face of the first set of rear wheels. This is in the aim to create an air curtain that reduces wake and air capture around the rear wheels. This slot remains the same width throughout the duct but reduces in height. It is aimed at the lower side of the rear wheel.

Concept Design 2



Figure 3.1.8: Detailed View of Trailer Skirt Inlet Geometry Design 2

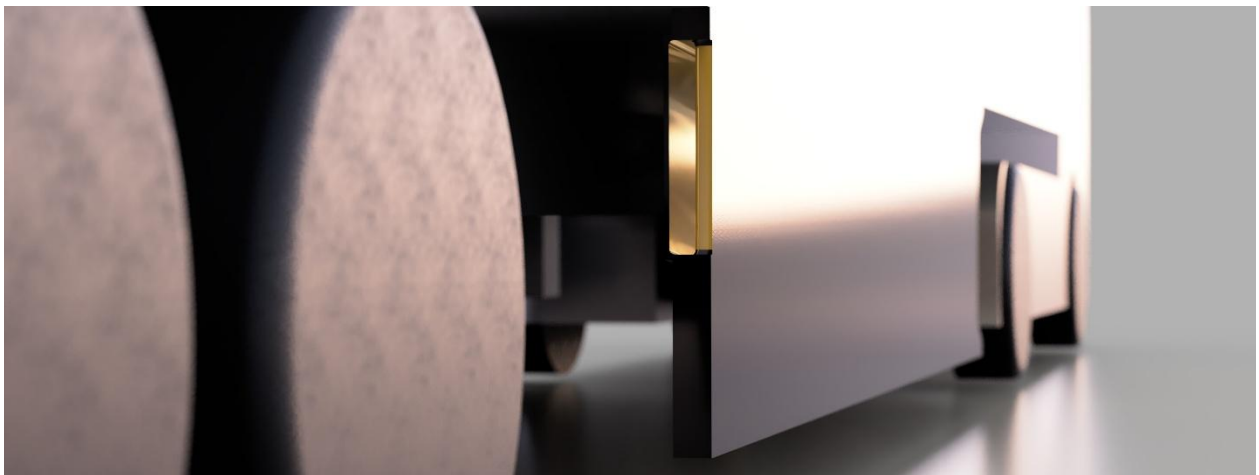


Figure 3.1.9: Detailed View of Trailer Skirt Outlet Geometry Design 2

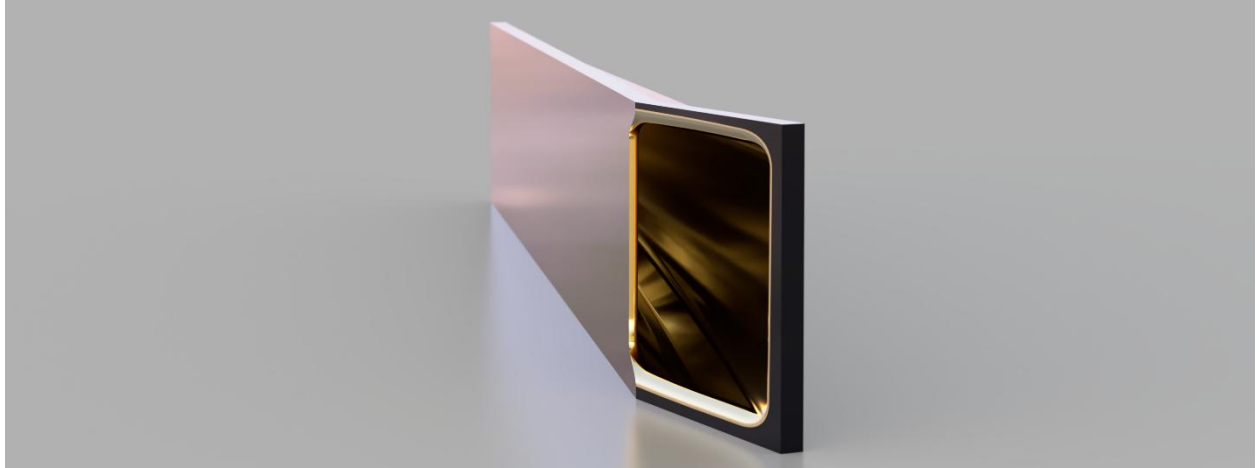


Figure 3.1.9: *Isolated View of Trailer Skirt Inlet Geometry Design 2*

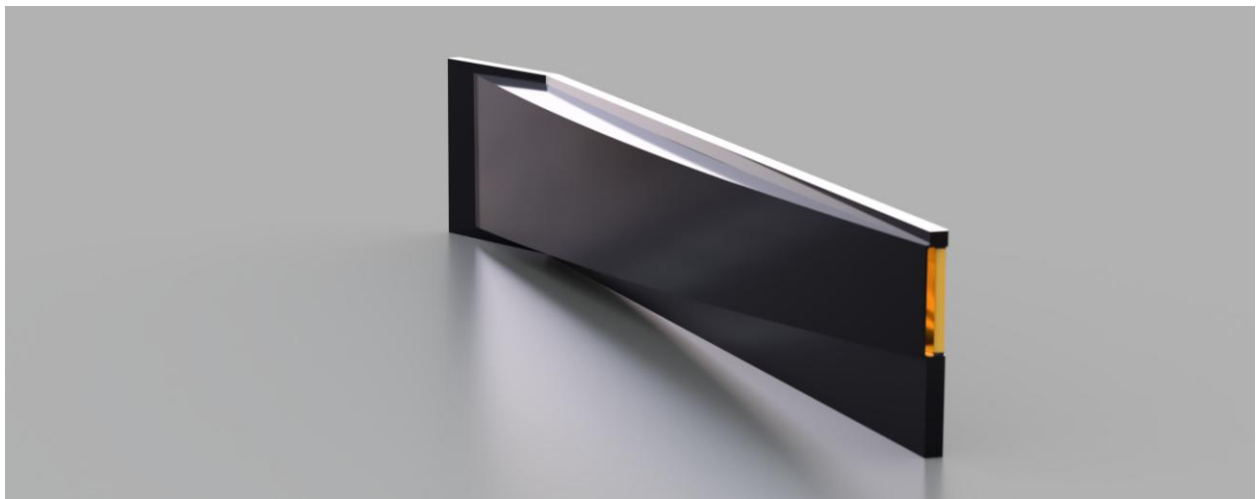


Figure 3.1.10: *Isolated View of Trailer Skirt Outlet Geometry Design 2*

Concept design 2 consists of a trailer skirt with a large opening on the leading face, aligned with the high-pressure zone found when testing. The principle of operation is similar to that of design 1. Notable changes that the duct entrance encompasses a much larger area, the duct is curved with fillets on the edges, to reduce friction and turning losses. The duct has a significantly larger entry to exit ratio, to extract maximum exit velocity from the high-pressure front region. The exit is aimed at the top of the wheel, and it tapers toward the edge. This design forgoes the full coverage over the wheel arches

3.2 Design Analysis

Designs 1 and 2 inlets, internal ducts, and outlet slots were modelled in SolidWorks 2024 and Autodesk Fusion 360, respectively, as a single skirt-mounted assembly. A defeatured tractor unit was included in both designs to capture wake effects on the trailer leading edge. Table 3.2.1 presents the design configurations.

Table 3.2.1: Duct Design Configurations

Parameters	Design 1	Design 2
Inlet Dimensions (mm)	830.23 x 129.71	850 × 850
Outlet Dimensions (mm)	268.86 x 90.3	50 × 400
Inlet Angle to Flow Direction	14.74°	18.5°
Duct Length (mm)	1343.24 + 4523.67	5000
Area Ratio	4.436:1	36.1: 1

The aerodynamic performance of design 1 and 2 was evaluated based on several key factors, including geometry shape, surface coverage of underside, ground clearance and airflow management.

Design 1 Aerodynamic Performance:

Design 1 incorporates a combination of flat panels with tapered edges within the trailer skirt to allow controlled airflow through the structure. The tapered geometry helps to guide airflow along the trailer surface, reducing flow separation (Hucho, 1993). The inclusion of narrow slim openings contributes to controlled airflow passage, maintaining a more streamlined flow compared to a fully solid skirt. Furthermore, design 1 provides significant coverage along the lower side of the trailer, enclosing front underside trailer and wheel regions. This reduces the amount of accelerated flow entering beneath the trailer, which is a major source of turbulence and drag. Ground clearance for design 1 is positioned close to the ground to minimise underbody airflow, this lowers the wake formation; however, it may introduce practical challenges such as an increased risk of contact with uneven road surfaces. This results in a trade-off between real world feasibility and aerodynamic performance.

Design 2 Aerodynamic Performance:

Design 2 introduces a revised trailer skirt geometry intended to increase air flow capture and deliver a higher momentum air curtain to the wheel region. This concept uses a significantly larger inlet opening on the leading face of the skirt, positioned on the leading face to capture incoming inflow. By increasing the inlet area, the design captures a greater volume of ram air, providing more available dynamic pressure for acceleration through the duct. The internal duct geometry has been reshaped into a smooth, curved profile with filleted edges to minimise internal friction and turning losses, helping maintain attached flow and a more uniform velocity

distribution at the outlet. The duct narrows from a wide inlet to a smaller outlet to accelerate the airflow, functioning as a passive nozzle that increases jet momentum at the discharge. The outlet is angled toward the upper region of the wheel, targeting the area where wheel-induced turbulence is strongest, and is positioned further outward to help seal the gap between the skirt and the external flow. Compared to Design 1, this configuration does not fully enclose the wheel arch, relying instead on the higher-momentum jet to provide the necessary shielding effect.

Design Comparison: Narrow Slot and Wide Curved Slot

The backflow in Design 1 (thin, narrow slot) under low driving-pressure conditions highlights an important trade-off. The high contraction ratio of the slot means the duct requires sufficient pressure head to overcome the exit velocity requirement and all upstream losses. When the available driving pressure is too low, the outlet becomes vulnerable: external wake pressure can exceed the internal duct pressure, reversing the flow direction. Design 2, has a wider outlet and lower exit resistance, is more robust under low-intake-pressure conditions, although this comes at the cost of a less concentrated discharge and a potentially weaker curtain effect when flow does occur.

This behaviour suggests that the two designs may be complementary across the yaw range. The narrow-slot configuration may perform more effectively and remain stable at low yaw angles, while the wider-outlet design, with its higher potential discharge velocity, may outperform at higher yaw angles where intake pressure is sufficient to drive flow through the more restrictive geometry. The complete drag-coefficient data across all yaw angles will determine whether this hypothesis is supported.

3.3. CFD Setup

A test matrix of four configurations at three yaw angles yields twelve independent simulations.

Table 3.3.1: CFD Test Matrix

No. of Runs	Configuration	Yaw angles
4	Baseline 1 (No skirt)	0°, 7.5°, 15°
	Baseline 2 (With Skirt)	
	Skirt + Design 1	
	Skirt + Design 2	

Yaw angles: 0°, 7.5°, 15°. The 7.5° case matches the average effective yaw ($\approx 7.1^\circ$) measured on UK HGVs (Barden and Gerova, 2016); 15° represents an extreme crosswind.

Solver and turbulence model: ANSYS Fluent, steady-state RANS with the k- ω SST model (Menter, 1994). Pressure-velocity coupling used the SIMPLE algorithm.

Geometry Design

All trailer configurations were prepared using simplified CAD models that retained the dominant aerodynamic features while removing small component details such as bolts, small fillets and small features. These features have minimal influence on the global flow field but significantly increase mesh complexity due to the size of the trailer. All models were exported as clean CAD models with no broken surfaces and unintended gaps, ensuring they could be meshed reliably.

Computational Domain

A wind tunnel approach was used for the computational domains to ensure realistic external flow development around the trailer. The inlet was positioned 30 m upstream of the HGV, with the outlet 90 m downstream, and the domain extended 40 m in width and 20 m in height. This corresponds to approximately five trailer lengths upstream and ten downstream, providing sufficient space for the wake to develop while maintaining a low blockage ratio of 1.25%. Symmetry was not applied because the wheel wake and the skirt's outlet jet introduce natural asymmetric flow structures, meaning the left and right sides of the domain do not behave identically.

A uniform inlet velocity of 30 m/s^{-1} with a turbulence intensity of 5% was applied, and a pressure outlet boundary condition was assigned at the downstream face. All vehicle surfaces were modelled as no-slip walls, meaning the air velocity at the surface was set to zero to accurately capture boundary layer development and surface friction drag. The ground was modelled as a moving wall with a speed of 25 m/s^{-1} and a surface roughness height of $k_s = 0.001 \text{ m}$ to better represent road conditions. Yaw angles were introduced by rotating the inlet velocity vector relative to the trailer orientation. **Figure 3.3.1** shows the domain setup values used for all models on Ansys.

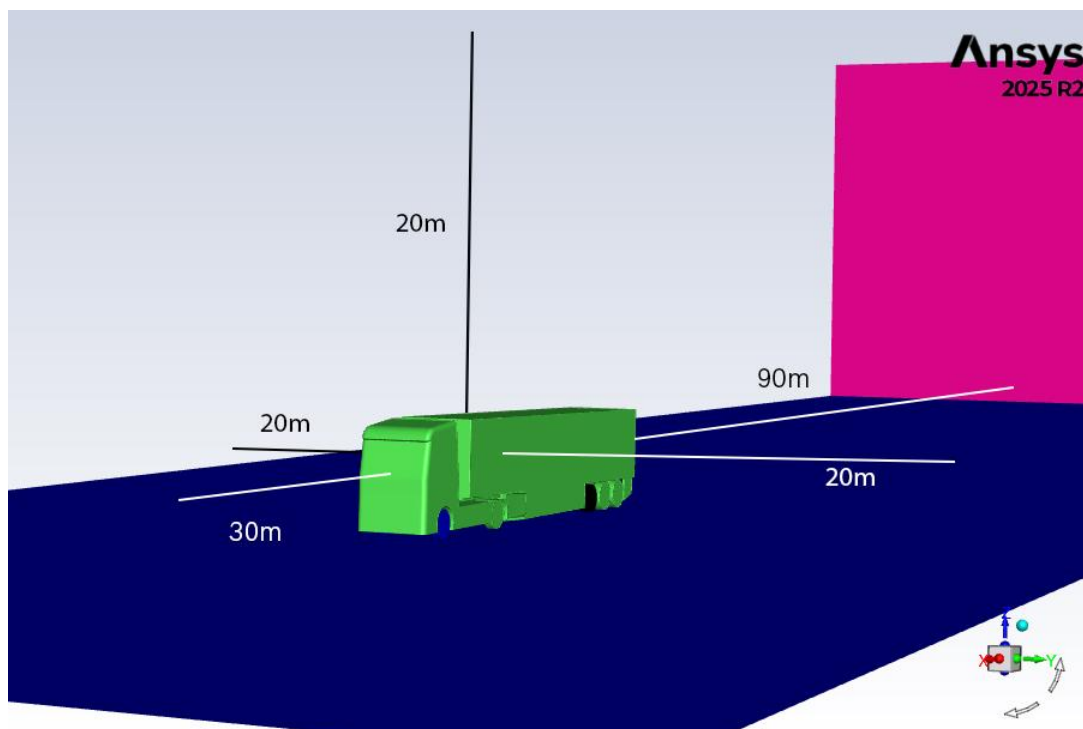


Figure 3.3.1: CFD Domain Dimension and Boundary Condition Setup

Validation

The models were validated against published drag co-efficient values for comparable HGV geometries.

3.3.1 Mesh and Boundary Conditions

A Hybrid unstructured mesh was generated, incorporating prism layers along all surfaces to achieve a target $y^+ < 1$. Local refinement was applied in critical flow regions, including the tractor-trailer gap, duct interior, outlet slot, and the bogie wake. A mesh-independence study demonstrated that the predicted drag coefficients changed by than 1 % beyond 5×10^6 cells.

3.4 System Integration Approach

A Systems Integration approach was used to manage the interfaces between the air curtain, the skirt, and the trailer. summarises the three critical interfaces, the design considerations applied, and the verification steps taken.

Aerodynamic	Intake located to capture pressure at zero yaw and windward-side dynamic pressure at yaw.	Full-vehicle CFD pressure contour analysis across yaw sweep.
	Intake lip and duct-to-skirt transition free of flow separation.	Surface pressure and wall-shear-stress plots along intake and duct.
Regulatory	Overall width ≤ 2.55 m (Construction & Use Regs 1986).	Dimensional check in CAD from vehicle centre-line to outer skirt edge.
	Ground clearance maintained; no rigid projections.	Minimum-height measurement at laden condition.
	Freedom to operate (no conflicting patents).	Prior-art patent search; results negative for skirt-integrated passive air curtains.

Table 3.4.1: System Integration Interface

By treating legal compliance as an engineering interface, alongside mechanical fit and aerodynamic function, the design process reflects professional practice in which regulatory requirements are design inputs.

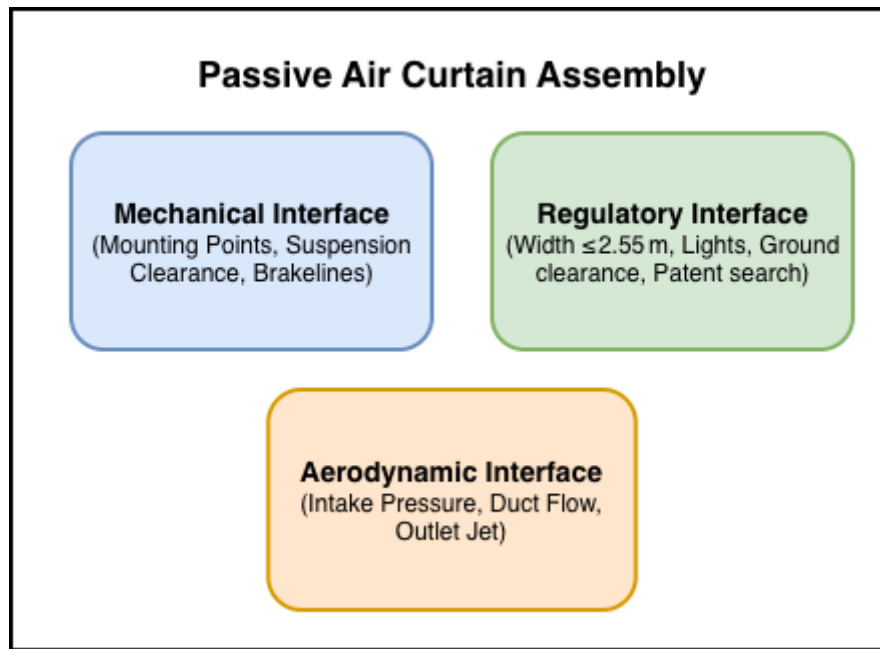


Figure 3.4.2: System Integration context diagram illustrating the aerodynamic and regulatory interfaces.

4. Results and Discussion

4.1 Design Comparison

Confuration	0° Yaw			7.5° Yaw			15° Yaw		
	Force	Cd	Cd reduction compared to baseline (%)	Force	Cd	Cd reduction compared to baseline (%)	Force	Cd	Cd reduction compared to baseline (%)
Baseline 1	2547.000	0.745	-	2602.000	0.761	-	2709.000	0.793	-
Skirted Baseline	2496.000	0.730	97.998	2546.000	0.745	97.848	2656.000	0.777	98.044
Design 1	2513.000	0.735	98.665	2562.000	0.750	98.463	2647.000	0.774	97.711
Design 2	2465.000	0.721	96.781	2524.000	0.738	97.002	2623.000	0.767	96.825

Table 4.1.1: Aerodynamic Peformance of all Configuraitons

Where:

$$\rho = \frac{1.225 \text{ kg}}{\text{m}^3} \quad (2)$$

$$Ux = 29.743 \text{ A} = 6.2\text{m} \quad (3)$$

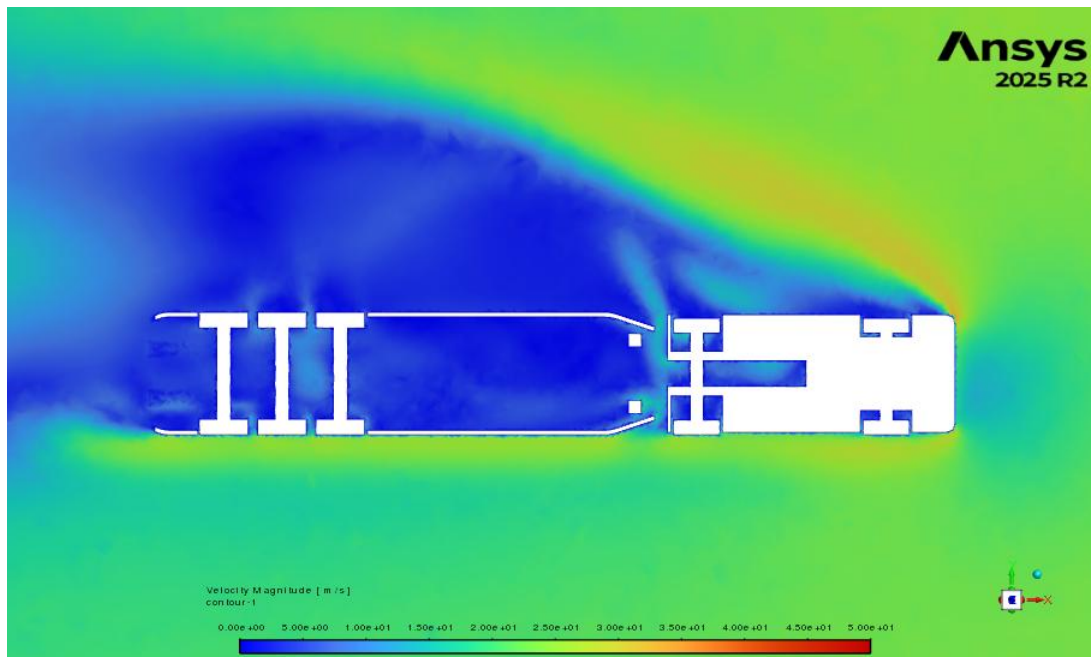


Figure 4.1.2: Velocity Contour Plot illustrating External Flow Behaviour Around Baseline 2

Figure 4.1.2 shows the velocity field illustrating largely uniform flow along the trailer sides, with local deceleration and inflow around the wheels. These wheel-induced disturbances generate small regions of reduced velocity that contribute to wake formation and increased turbulence in the lower side region.

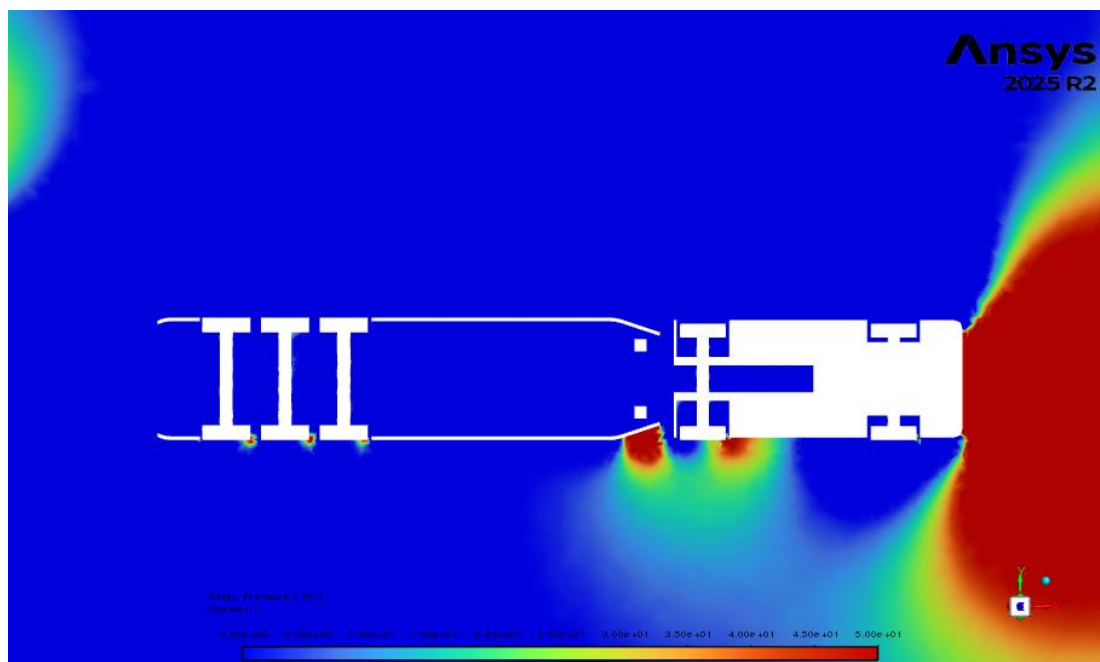


Figure 4.1.3: Pressure Distribution on Baseline 2 at 7.5° Yaw

The pressure contour highlights the asymmetric loading generated by the 7.5° yaw angle. A pronounced high-pressure region forms on the face of the skirt, indicating that this surface receives the majority of the oncoming flow impact, while lower pressure zones develop along the wake side as the flow accelerates and separates.

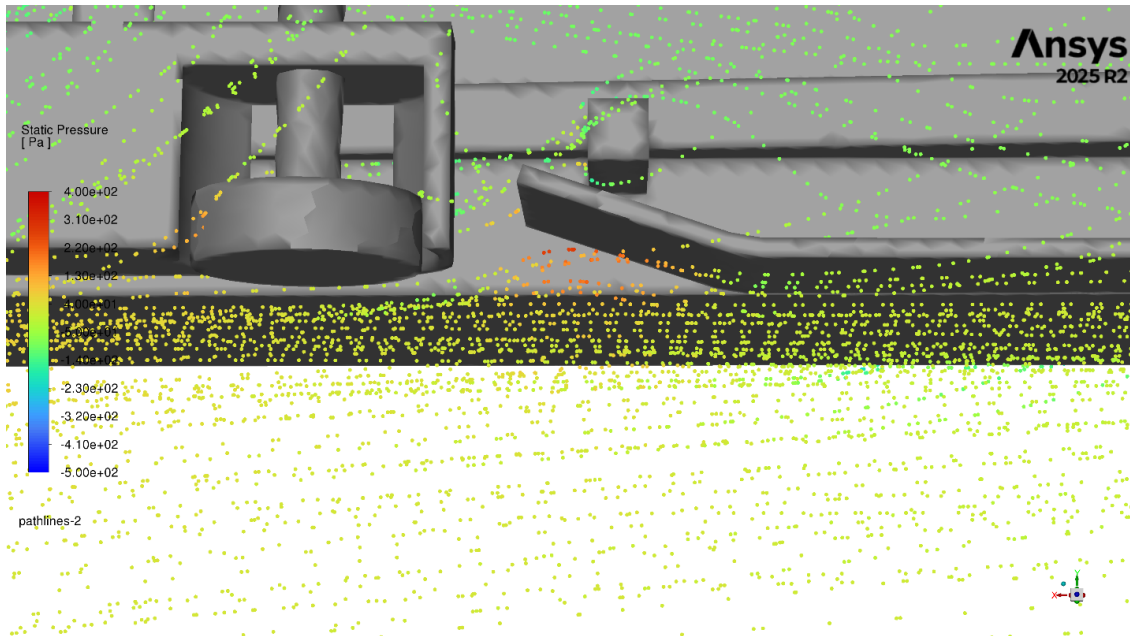


Figure 4.1.4: Flow trajectories with Static Pressure of Baseline 2

Figure 4.1.4 shows the flow trajectories, coloured by static pressure, highlighting the rise in pressure on the forward face of the side skirt as the incoming flow hits the surface. Lower pressure regions develop downstream as the flow accelerates and moves beneath the body.

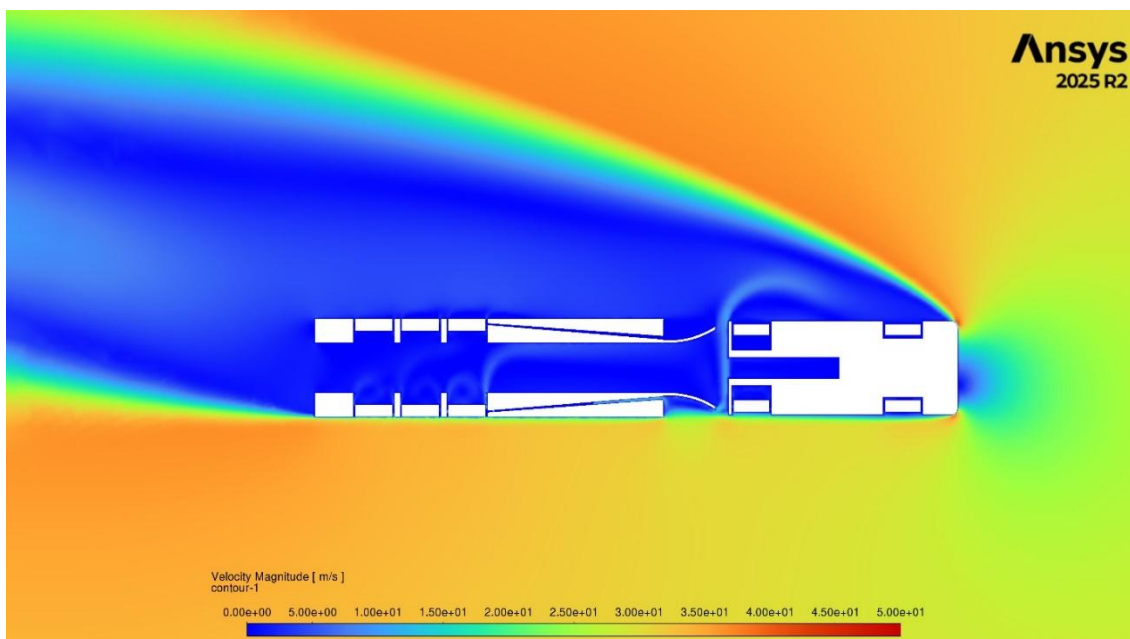


Figure 4.1.5: Velocity Contour Plot illustrating External Flow Behaviour Around Design 1

Figure 4.1.5 shows the velocity contour plot for Design 1, illustrating how the flow accelerates through the narrowed regions of the geometry and slows in the wider sections. The distribution

of high velocity zones indicates that the modified internal passages strongly influence the flow path and create pronounced acceleration. In contrast, the baseline 2 configuration exhibited mostly uniform external flow with far weaker velocity gradients. Compared with the baseline 2, Design 1 generates significantly higher peak velocities and a more distinct pattern of velocity changes, demonstrating that the modified skirt has a much greater effect on directing and accelerating the flow.

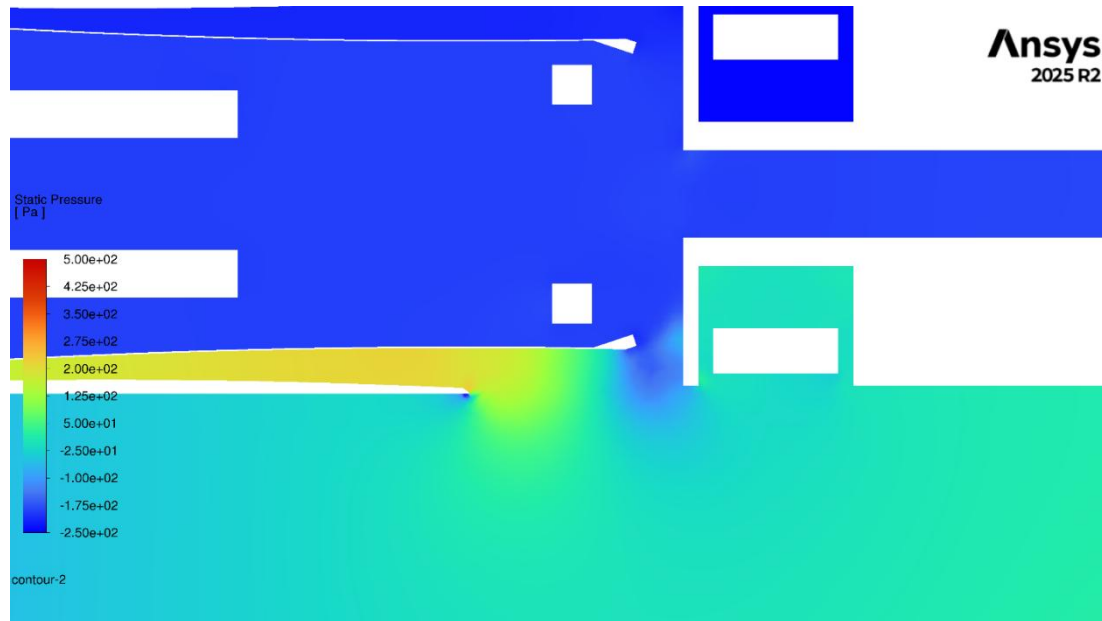


Figure 4.1.6: Design 2

Figure 4.1.6 shows the pressure contour plot for design 2, indicating a rise in pressure within the wider upstream section of the duct and a clear pressure drop as the duct narrows toward the rear. This behaviour reflects the expected relationship between duct geometry and static pressure, where flow acceleration through a constriction leads to lower pressure. In contrast to the baseline 2, which exhibited external pressure variations around the body, design 2 demonstrates a more controlled internal pressure distribution driven directly by the duct shape. This behaviour is consistent with the fluid flow theory, where a reduction in duct area causes the flow to accelerate and the static pressure becomes lower.

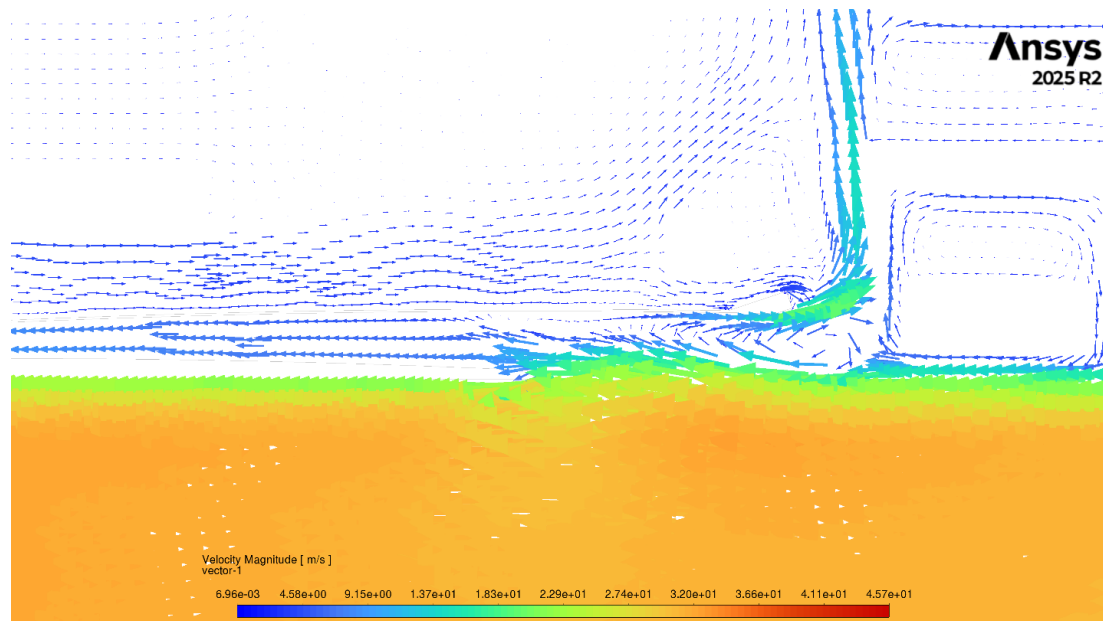


Figure 4.1.7: Velocity Contour Plot for Design 2

Figure 4.1.7 shows the velocity contour plot for design 2, illustrating how the flow accelerates as it moves past the geometry and decelerates in the recirculation region behind the obstruction. The contour field highlights clear velocity gradients, with higher speed flow developing along the unobstructed path and lower velocity zones forming in the wake. The velocity vectors also confirm that the flow continues through the ducting as intended, with no signs of backing up or stagnating within the passage. Compared with the baseline 2, which displayed smoother external flow with less pronounced separation, design 2 generates stronger variations in velocity due to the altered flow path.

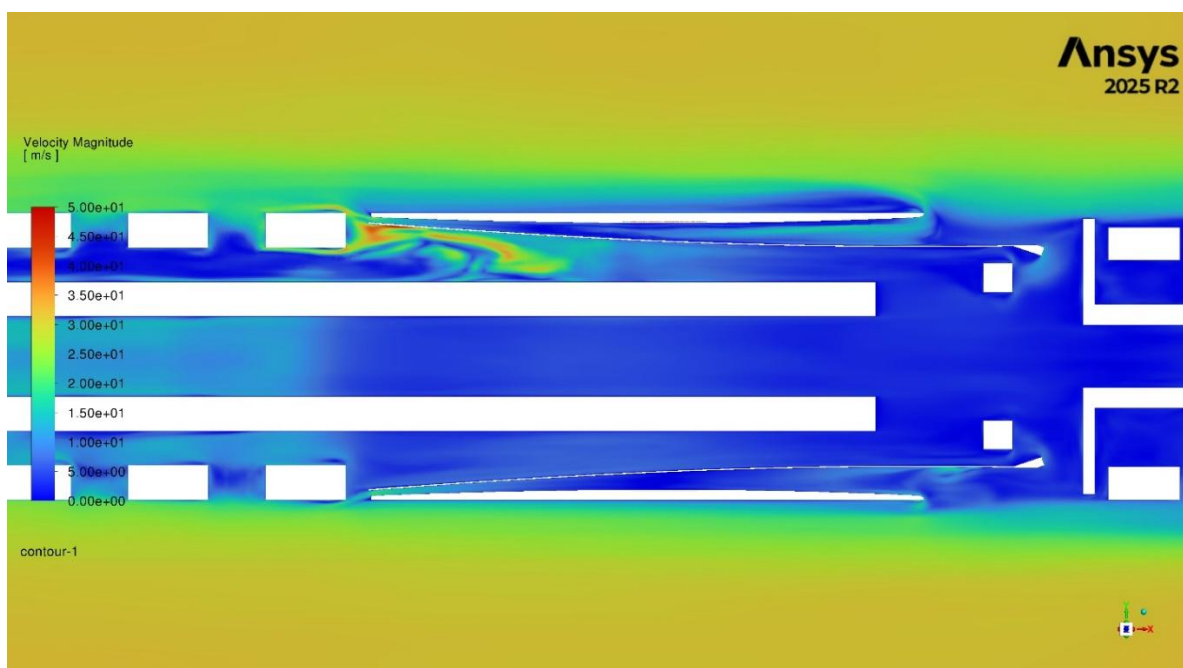


Figure 4.1.8: Velocity Contour Plot at 0° Yaw for Design 2

Figure 4.1.8 shows the velocity contour plot for design 2 at 0° yaw, where both ducts become active and draw flow through the side ducts. The velocity field indicates clear acceleration along the ducts, confirming that the flow is not stagnating or backing up within the geometry. Some turbulence develops on the inner side of the top skirt, caused by the interaction between the duct flow and the surrounding external stream. Compared with the baseline 2, which exhibited smoother external flow and no internal duct activity, design 2 demonstrates a more complex velocity distribution driven by the ducting system and the modified skirt geometry.

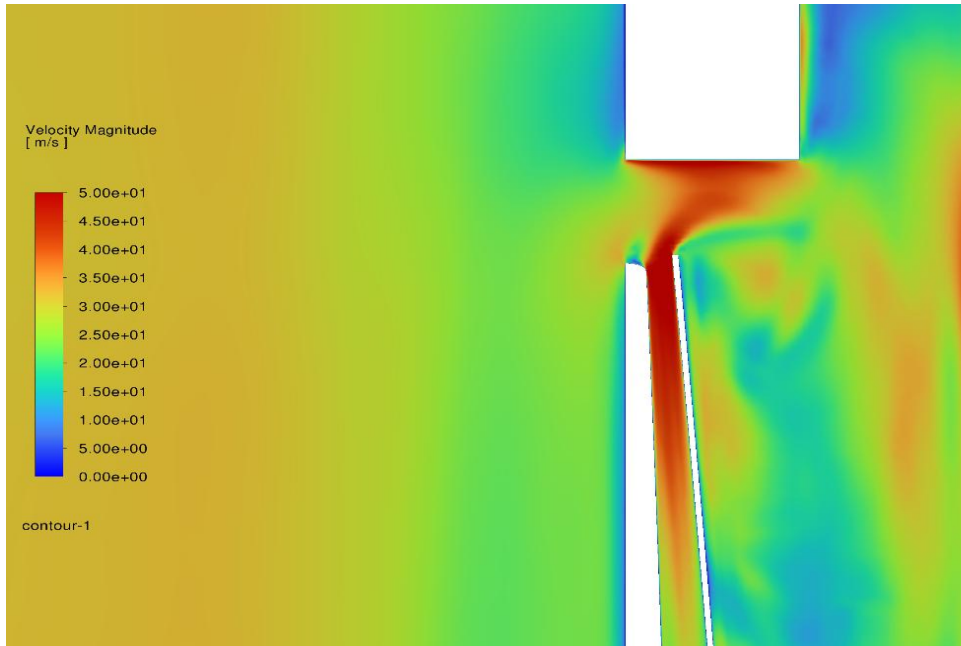


Figure 4.1.9: Velocity Contour Plot at 7.5° Yaw for Design 2

Figure 4.1.9 shows the velocity contour plot for design 2 at a 7.5° yaw angle. The duct continues to accelerate the flow, producing a jet with approximately double the free stream velocity. However, the outflow trajectory becomes misaligned under yaw, causing the jet to hit directly on the wheel rather than being directed cleanly downstream. This interaction generates additional turbulence and indicates that the duct's exit geometry does not provide adequate control of the jet direction at off-axis flow conditions. Compared with the baseline 2, which exhibited no jet and wheel interaction, design 2 introduces a new aerodynamic issue that may negatively affect drag and downstream flow behaviour.

4.2 Limitations

The study faced several practical and computational constraints that influenced the modelling approach and the accuracy of the results. The computational cost of generating a sufficiently refined 3D mesh, made full 3D simulation impractical within the project timeframe. As a result, a 2D approach was adopted to allow consistent meshing, faster convergence, and reliable comparison between design iterations. The 2D approach allowed consistent meshing, faster convergence, and reliable comparison between design iterations, but it inherently removes 3D

flow structures such as vortices forming in the wheel wake. As a result, the findings are interpreted as representative trends rather than absolute aerodynamic values.

Mesh resolution was also constrained by available computational resources. While refinement was applied in key regions such as the ducts, wheel area, and wake, a fully resolved boundary layer mesh was not achievable across the entire domain. This may affect the accuracy of separation prediction, near-wall velocity gradients, and local turbulence intensity. A finer mesh or adaptive refinement would likely improve accuracy but was not feasible within the available hardware and time.

Boundary conditions and geometric simplifications also affect the realism of the results. The inlet velocity was uniform, whereas real world wind conditions are turbulent and variable. Ground interaction was simplified, and wheel rotation was not modelled, which may influence the accuracy of flow behaviour around the lower body. Some geometric features were idealised to reduce meshing complexity, meaning the model does not fully represent manufacturing detail or surface roughness.

5. Conclusion

Preliminary Findings

CFD simulations are ongoing. The following preliminary observations have been made from initial contour plot analysis of velocity and pressure fields on representative cross-sectional planes through the air curtain duct and bogie region. These findings are qualitative and will be supplemented by full drag coefficient data when the complete run matrix is available.

Intake Pressure Distribution

Pressure contour analysis on the skirt side surface reveals that, at zero yaw (0°) and moderate yaw (7.5°), the leading skirt intake region receives limited positive gauge pressure. The skirt side of an HGV trailer at low yaw is a largely neutral-to-slightly-positive C_p region; it is not a primary stagnation zone, and the oblique orientation of the intake at 18.5° to the flow further reduces the effective ram pressure at the inlet face. This is consistent with expectations from pressure budget theory: the available intake driving pressure at low yaw is modest.

At 15° yaw, however, the windward skirt face experiences a substantially increased pressure. Preliminary contour data indicates a pressure differential of approximately 1,200 Pa between the high-pressure windward intake region and the low-pressure leeward reference, representing a significant passive driving force for the air curtain system. This value exceeds the freestream dynamic pressure of ~551 Pa at 30 m/s, suggesting that local flow stagnation and curvature effects at the yawed leading face amplify the available intake pressure beyond simple dynamic pressure recovery. This finding implies that the air curtain concept may be particularly effective under crosswind conditions, where the intake pressure is strongest.

[Figure 3: Pressure contour on skirt side surface — 0° , 7.5° , and 15° yaw comparison [INSERT WHEN AVAILABLE]]

Duct Flow Quality

Velocity contour planes within the air curtain duct show that, under the low-yaw conditions where intake pressure is limited, the duct internal velocity is correspondingly low. In the thin narrow slot configuration (Design B), initial analysis shows evidence of localised backflow within the duct under certain conditions. This is consistent with an insufficient pressure differential to drive stable, unidirectional flow through the narrow geometry: when the driving pressure is low relative to the resistance of the outlet slot and duct, the system does not maintain a net positive flow direction, and reverse flow can enter through the slot from the external low-pressure wake region.

This backflow phenomenon does not appear to be present in the wide curved slot configuration (Design A) under the same conditions, suggesting that the lower resistance of the wider outlet allows the available pressure to maintain forward flow more readily. This is an important preliminary distinction between the two design variants and underscores the sensitivity of passive system performance to the pressure budget balance between intake and outlet geometry.

A proposed improvement of our design is the incorporation of a sensor-controlled rear duct system. It would work as an active aerodynamic extension of our pre-existing system to further reduce drag. This concept builds on the logic of our system by replacing the passive rear treatment with a ducted, sensor-informed outlet system. It works as a form of controlled base bleeding. The introduction of a higher-pressure air into a negative pressure region behind the vehicle, reducing pressure differential and therefore drag. “Active base bleeding uses power air supply and passive base bleeding, which ducts naturally occurring high-pressure air to low-pressure regions.” (Patten et al. 2012) This concept would now incorporate both ideas by passively capturing ram air through the front and upper-side intakes, whilst also adding active sensing, as well as outlet modulation at the rear. Within the context of a practical setup, twin upper-side ducts would feed a rear manifold and an adjustable slot outlet, which would find itself above the trailer doors. The controller would then modulate the outlet area in accordance with the rear pressure conditions and the vehicle’s speed. The graph below shows how the system would have a threshold for both speed and pressure, activating the system when they conjoin at the activation threshold. The activation region can be denoted by the grey discoloured region of the graph.

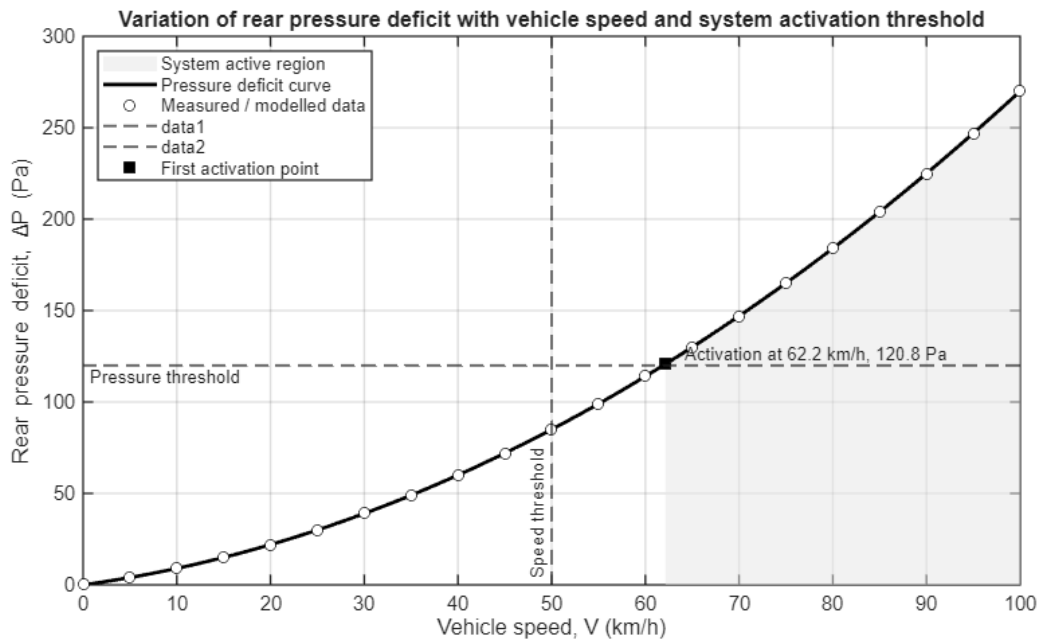


Figure 5.1: MATLAB graph conceptualising a pressure deficit curve, showing that when speed of the vehicle increases, the rear pressure deficit increases with it.

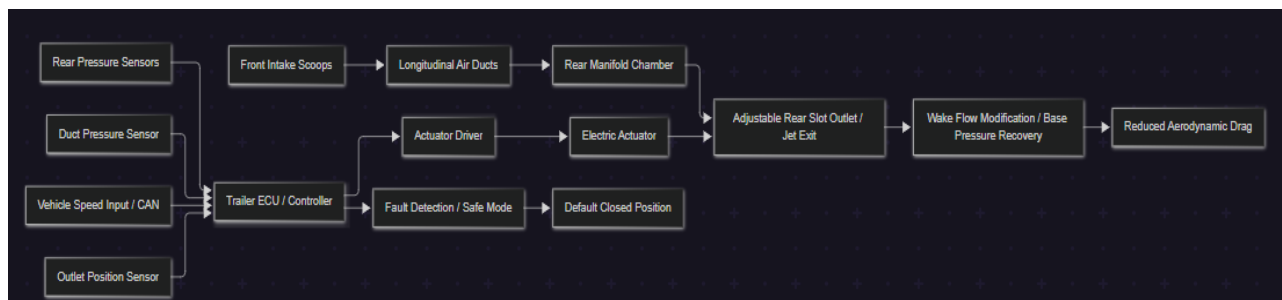


Figure 5.2: Block diagram showing how the sensor-controlled system uses rear pressure, duct pressure, vehicle speed and outlet position to control an actuator-driven rear slot outlet. This is done with the objective of reducing aerodynamic drag.

The above block diagram shows the organisation of the system, how information and airflow move through it. Air enters through the front intake scoops, then proceeds to pass through the longitudinal ducts, collecting in the rear manifold chamber, before eventually leaving out of the adjustable rear slot outlet. That is the overall airflow path. Alongside the airflow path there is also the control path. Rear pressure sensors measure the pressure conditions at the back of the trailer. The duct pressure sensor will then check whether enough airflow is available inside the duct. The vehicle speed input will then tell the system whether aerodynamic control should be activated alongside the outlet position sensor, which confirms whether the rear outlet has opened to the dictated position. All these signals are then sent to the controller (ECU), which will then make the according decisions. The ECU will then send information to the actuator driver, to power the electric actuator. This actuator will physically adjust the rear slot output, optimising the way air is discharged into the wake. The final 3 blocks then act as the intended outcomes of the

system due to the actions of the ECU. The fault detection system is proof that the ECU would both attempt to improve aerodynamics and monitor problems. This allows it to denote whether a sensor has failed, or the actuator is not responding correctly, telling the system to revert to the default closed position, so the trailer can behave like a normal trailer safely.

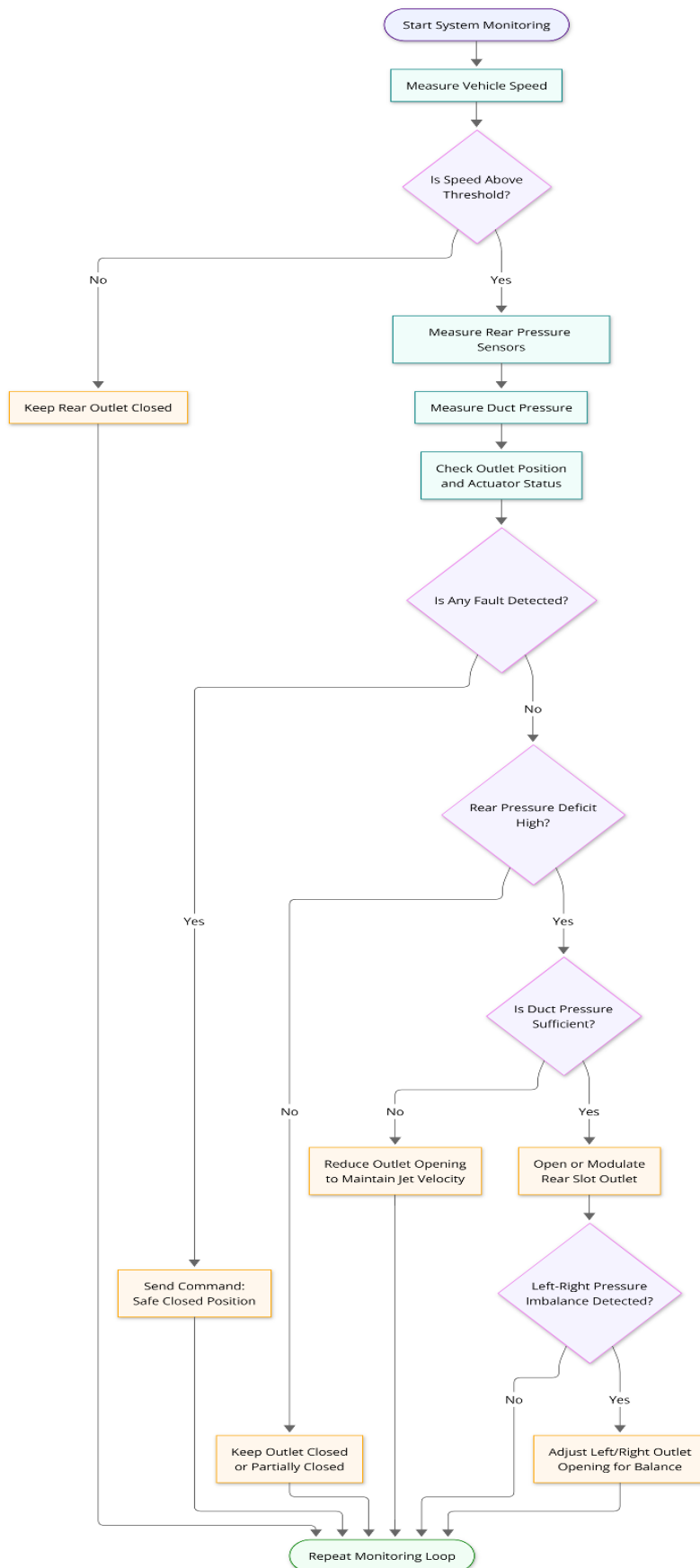


Figure 5.1: Flowchart diagram showing how a sensor-controlled system logically, step by step decides how to affect outlet control to help the trailer deal with aerodynamic stress.

Figure 5.3, explains the decisions that the controller will make in sequence during the operation. The system is started and begins by monitoring the speed of the vehicle. This is done as the system only needs to operate at high speeds when drag is significant, if speed were to be below the threshold the outlet would stay closed, continuing to monitor. In a situation where speed is high enough, the controller reads the status of the rear pressure sensors, duct pressure and outlet position/actuator status, as well as checking for any fault conditions. If a fault is found the system immediately enters the safe closed position for the outlet to protect from damage. Assuming there is no fault occurring, the controller will ask if the rear pressure deficit has reached the threshold to justify its intervention. If wake deficit is high, the controller then proceeds to check if the duct pressure is sufficient. This is important because if the outlet is opened fully, there needs to be enough air available to produce a meaningful jet that can be useful. If the duct pressure is at an adequate level, the controller will then proceed to modulate the rear slot outlet, opening it. Afterwards, performing another refinement by checking for any left to right pressure imbalance. This allows the system to check for any asymmetrical wake behaviour from things such as crosswinds. If the pressure is unbalanced, the controller will bias towards whichever side it needs to compensate for the imbalance. In totality, the logic diagram accurately depicts how the system is always working on condition-based reactions rather than always being active. It will only operate when specific thresholds have been met, including safety fallbacks essential for safe practical trailer-systems. Conclusion

This report has presented the design, theoretical basis, and CFD evaluation framework for a passive air curtain integrated into an HGV trailer side skirt. The device is motivated by the demonstrated effectiveness of wheel air curtains on modern passenger vehicles and by the established importance of the trailer bogie region as a primary source of HGV aerodynamic drag. The system requires no external power, relying on an 850 mm × 850 mm intake angled at 18.5° to the flow to recover dynamic pressure and route it through a 5 m duct before discharging through a 50 mm × 400 mm slot.

Theoretical analysis confirms that, at the simulation speed of 30 m/s, approximately 551 Pa of freestream dynamic pressure is available. The high inlet-to-outlet area ratio of 36:1 provides significant velocity amplification potential at the slot, subject to adequate pressure recovery at the intake. The 5 m duct length, with its large cross-section, contributes to minimal friction losses, making the intake location and slot geometry the dominant design parameters.

Preliminary CFD findings highlight significant yaw dependence. At zero and low yaw, the intake region receives limited positive pressure, constraining curtain performance. At 15° yaw, a pressure differential of approximately 1,200 Pa is observed, substantially exceeding the freestream dynamic pressure and suggesting enhanced curtain driving force under crosswind conditions. The narrow-slot design (Design B) exhibits backflow at low-pressure conditions, while the wide curved slot (Design A) appears more robust. The choice of 7.5° as the representative yaw angle is supported by UK on-road HGV measurement data indicating an average effective yaw angle of approximately 7.1°.

Quantitative drag coefficients and final design recommendations will be presented upon completion of the full 12-simulation run matrix. The preliminary results indicate that the concept is most promising as a crosswind drag-reduction device, and that intake geometry optimisation for improved low-yaw pressure recovery may be an important direction for design iteration.

6. References

1. Bayerische Motoren Werke Aktiengesellschaft (BMW), n.d. *EfficientDynamics: Aerodynamics – More Agility. Less Drag.* [Online]
Available at: https://www.bmwmyanmar.com/en_MM/topics/fascination-bmw/efficient-dynamics/aerodynamics.html
[Accessed 25 April 2026].
2. Cooper, K. R., 2003. *Truck Aerodynamics Reborn - Lessons from the Past.* s.l., SAE International.
3. Hucho , W.-H., 1993. *Aerodynamics of Road Vehicles.* s.l.:Annual Reviews Inc..
4. Hyundai , 2024. *Technology & Innovation.* [Online]
Available at: <https://www.hyundai.com/worldwide/en/newsroom/detail/hyundai-motor-and-kia-unveil-%25E2%2580%2598active-air-skirt%25E2%2580%2599-technology-to-help-electric-vehicles-go-faster-and-farther-0000000403>
[Accessed 20 April 2026].
5. K. Gerova & J. Barden, 2016. An on-road investigation into the conditions experienced by a heavy goods vehicle operating within the United Kingdom. *Transportation Research Part D*, Volume 48, p. 5.
6. Nabutola, K. L. & Boetcher, S. K. S., 2021. Assessment of conventional and air-jet wheel deflectors for drag reduction of the DrivAer model. *Advances in Aerodynamics*, 3(1), p. 29.
7. North American Council for Freight Efficiency (NACFE), 2016. *Annual Fleet Study*, s.l.: North American Council for Freight Efficiency.
8. North American Council for Freight Efficiency, 2015. *Trailer Aerodynamics Confidence Report* , s.l.: NACFE.
9. The North American Council for Freight Efficiency (NACFE), 2024. *How Trailer Aerodynamic Upgrades Deliver Over 10% in Fuel Savings.* [Online]
Available at: <https://www.kalestruck.com/buying-guides/how-to-utilize-aerodynamics-to-improve-fuel-efficiency/>
[Accessed 20 April 2026].
10. Transport Canada, 2012. *Review of Aerodynamic Drag Reduction Devices for Heavy Trucks and Buses*, Ottawa: Transport Canada.
11. White, F. M., 1999. *Fluid Mechanics.* 4th ed. s.l.:McGraw-Hill.

7. Appendix

Pressure Budget

The net slot exit pressure above ambient is the difference between the intake pressure recovery and all internal losses:

$$\Delta p_{\text{slot}} = \Delta p_{\text{intake}} - \Delta p_{\text{duct}} - \Delta p_{\text{exits}}$$

The intake pressure recovery (Δp_{intake}) depends on the pressure coefficient C_p at the intake location and the local dynamic pressure. The intake is oriented at 18.5° to the vehicle travel direction, which means it presents a partially oblique face to the oncoming flow. At zero yaw, this reduces the effective stagnation pressure at the inlet; at positive yaw on the windward side, the intake angle becomes more favourable and recovery may increase substantially. This yaw-dependent behaviour is one of the key phenomena the CFD study is designed to quantify.

Geometry and Area Ratio

The key geometric parameters of the air curtain system are summarised below:

Parameter	Value	Notes
Inlet dimensions	850 mm × 850 mm	Area = 0.7225 m ²
Inlet angle to flow direction	18.5°	Partial oblique face to oncoming flow
Duct length	5 m	Longitudinal, within skirt structure
Jet outlet dimensions	50 mm × 400 mm	Area = 0.020 m ²
Area ratio (inlet: outlet)	36.1 : 1	High contraction — velocity amplification potential

The area ratio of 36:1 between the inlet and the outlet slot is significant. In an idealised, lossless system with continuity, this ratio would imply a large velocity amplification at the slot. In practice, the exit velocity is limited by the available pressure budget: the slot cannot accelerate the flow beyond what the pressure differential across the duct can drive. The practical implication is that even modest pressure recovery at the intake, a fraction of the ~550 Pa freestream dynamic pressure, can drive useful slot exit velocities through this geometry. However, the high contraction ratio also means the system is sensitive to any reduction in inlet C_p , since a small pressure loss at the inlet translates directly into a proportionally large reduction in slot exit momentum.

Duct Friction Losses (Darcy–Weisbach)

Frictional pressure loss along the 5m duct is estimated using the Darcy–Weisbach equation:

$$\Delta p_f = f \cdot L/D_h \cdot (\rho V_d^2)/2$$

Where:

f is the Darcy friction factor,

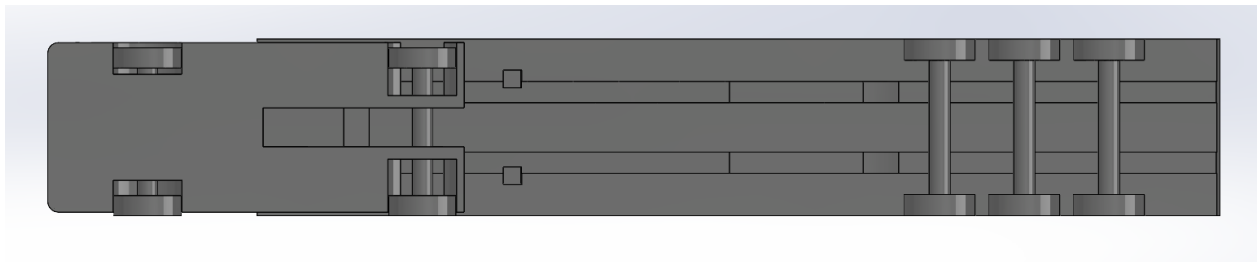
L is the Length of the duct; in this case 5m,

D_h is the hydraulic diameter of the duct cross section,

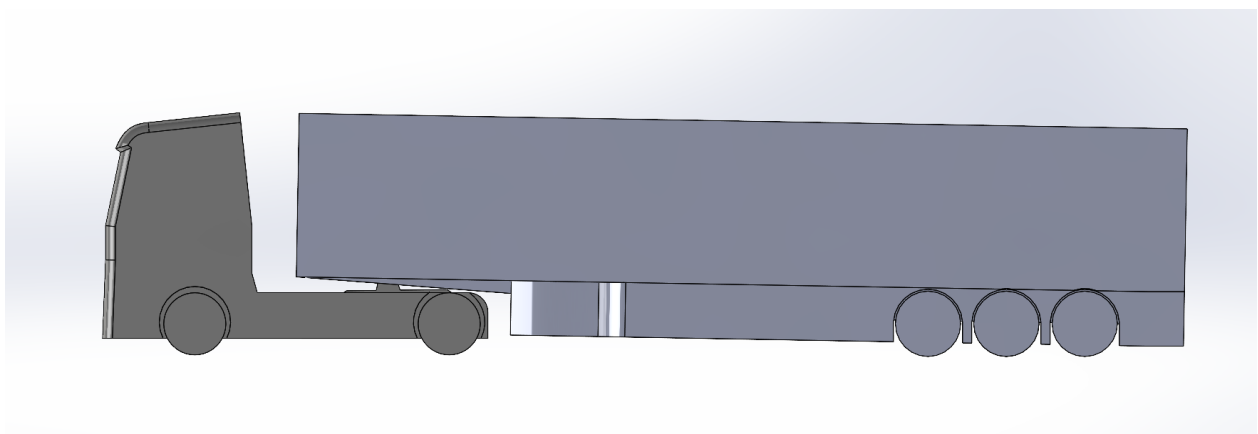
V_d is the mean duct velocity.

The duct maintains a large cross-sectional area (0.72 m^2 , matching the intake) along its length, giving a hydraulic diameter of approximately 0.85 m. Under the flow velocities likely to be encountered (on the order of a few metres per second to a few tens), the resulting pressure loss is estimated to be less than 50 Pa, which is small compared with the freestream dynamic pressure. The pressure budget is therefore dominated by the intake recovery and the exit conditions, not by duct friction.

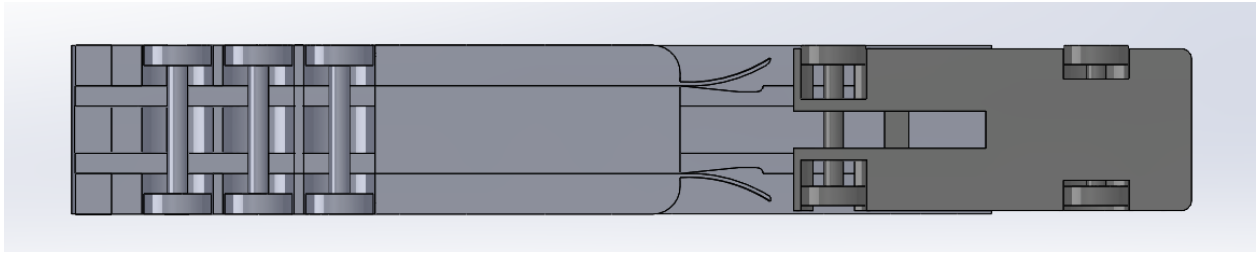
The very high area ratio has two important consequences. First, if the pressure budget is sufficient, continuity dictates a large velocity amplification from intake to outlet, which helps produce a high-momentum curtain. Second, the same contraction increases the outlet resistance: a small driving pressure may be unable to sustain forward flow, potentially leading to backflow through the slot.



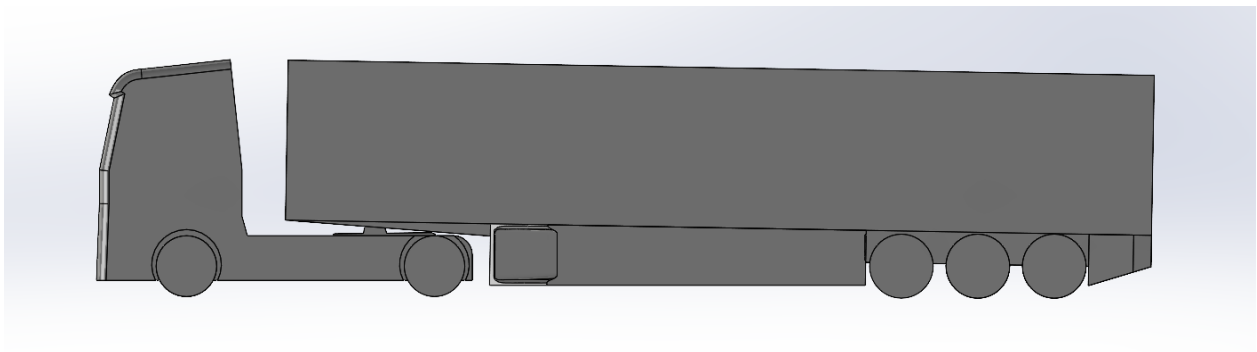
Bottom View of HGV Truck Without Trailer Skirt



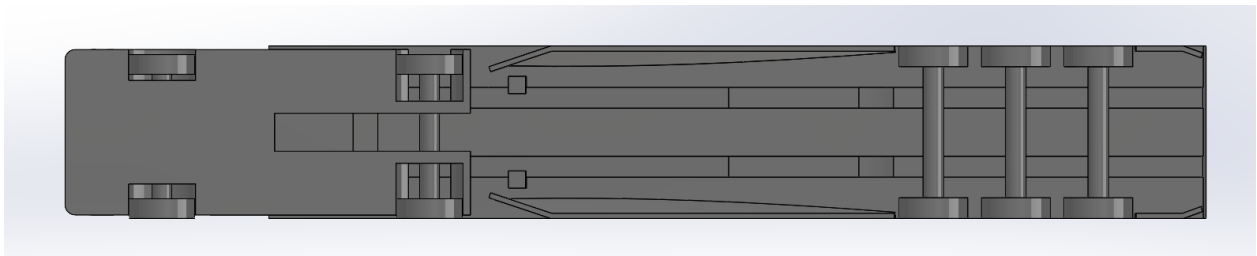
Side View of Trailer Skirt Design 1



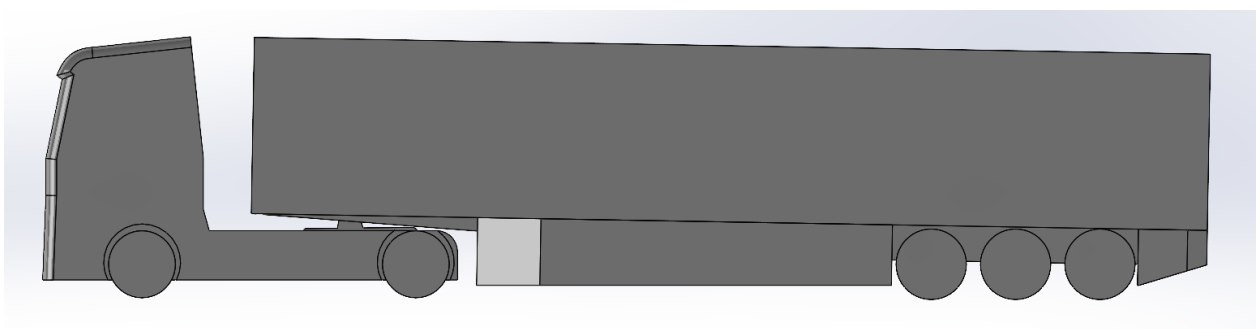
Bottom View of Trailer Skirt Geometry Design 1



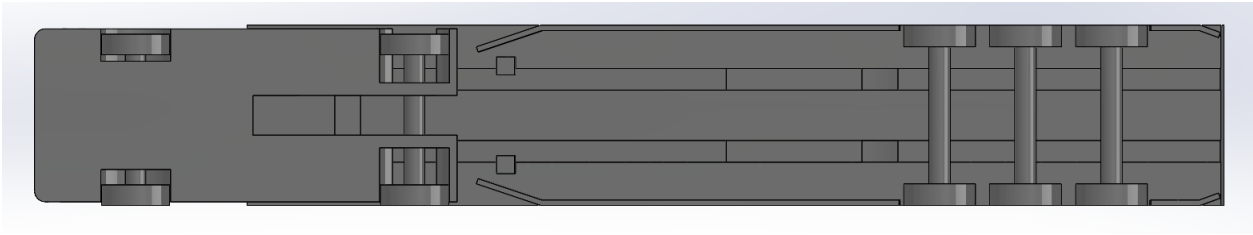
Side View of Trailer Skirt Design 2



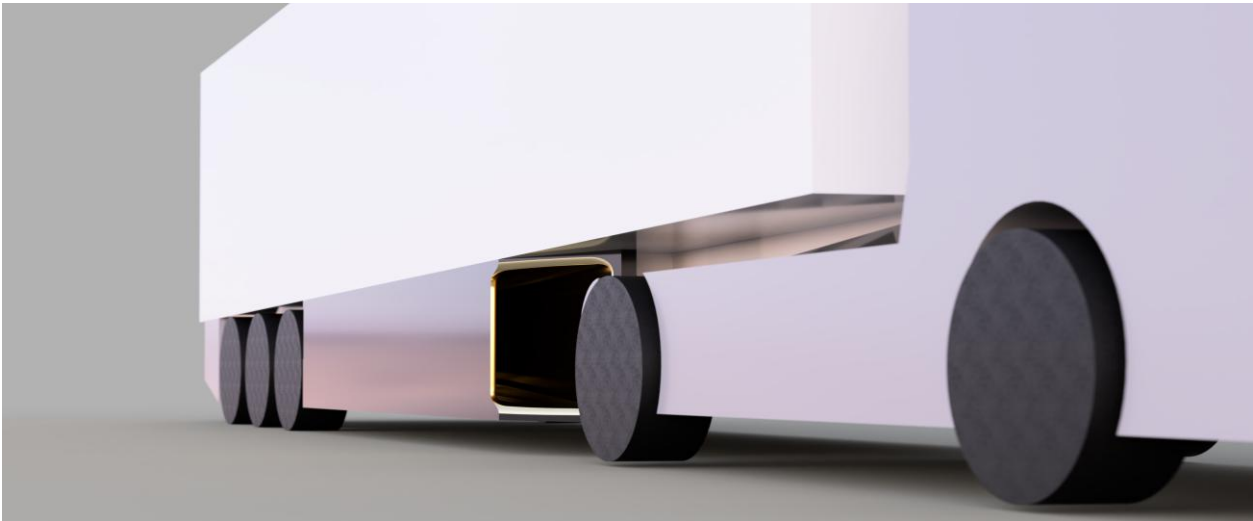
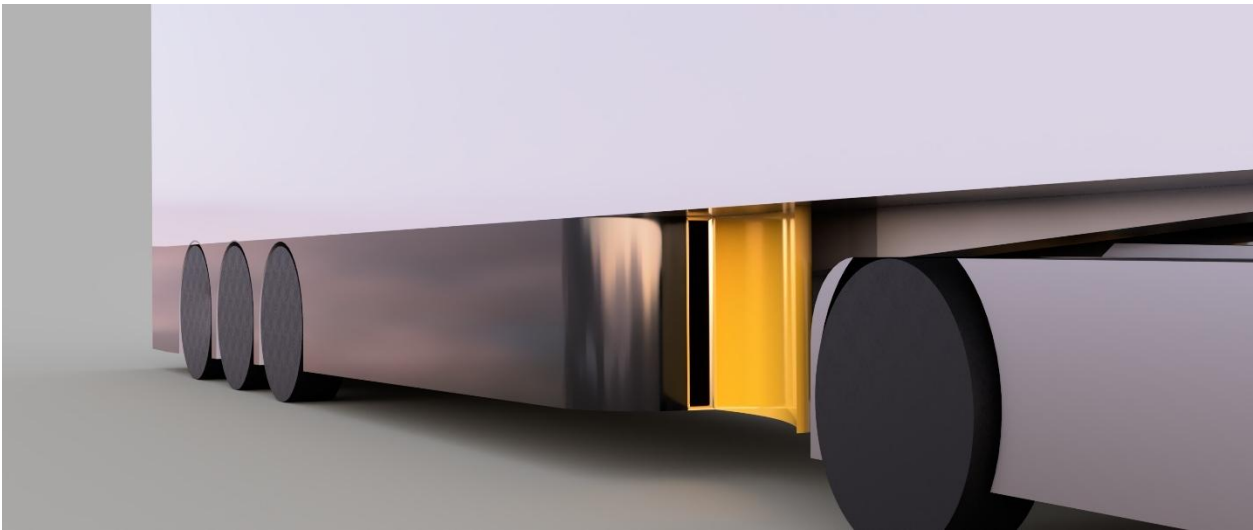
Bottom View of Trailer Skirt Design 2



Side View of Trailer Skirt Without Holes Design 3



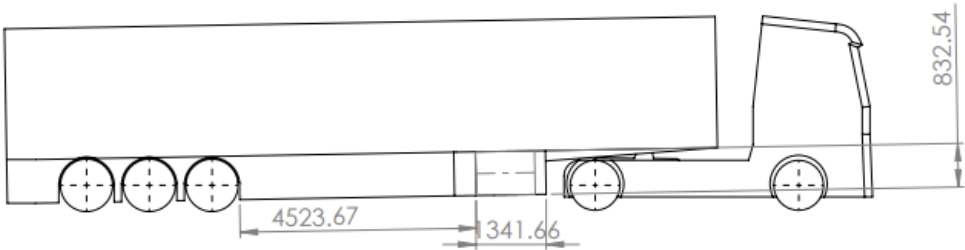
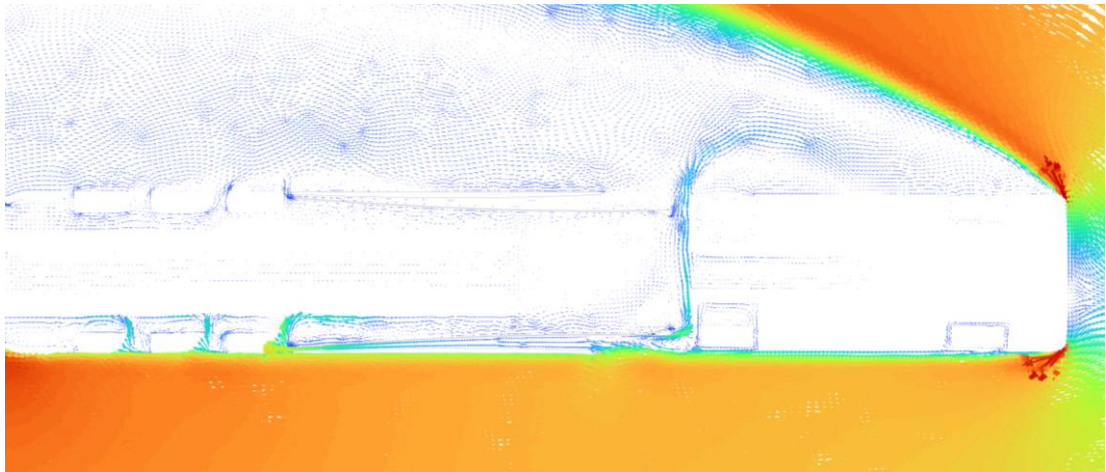
Bottom View of Trailer Skirt Without Holes Design 3



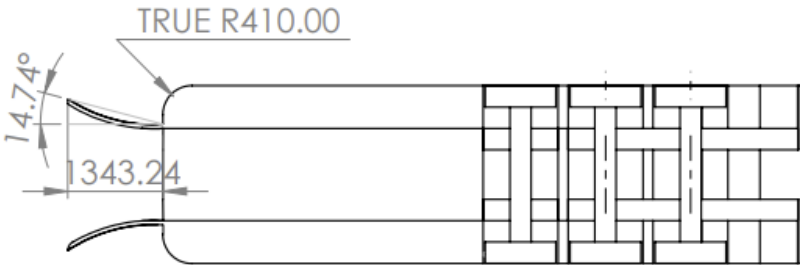
Interface	Design considerations	Verification / Validation
Mechanical	Mount to existing skirt-fastener points on the trailer side rail.	CAD clash detection and assembly checks.
	Clear full suspension travel, steering articulation, brake callipers, air lines, and ABS wiring.	Suspension-travel clearance study in SolidWorks.

	Breakaway fasteners to ensure safe separation in impact.	Fastener shear-out calculation.
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Systems Integration: Future Work



Design 1. SolidWorks Side View Drawing



Design 1 SolidWorks Bottom View Drawing